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VisionAid: Prompting Strategy and Confidence Calibration for Small Quantized Vision-Language Models on Degraded Product Photographs

***Abstract*—*Small, quantized, open-weight vision-language models (VLMs) are a realistic route to affordable, offline assistive reading for blind and low-vision people. Whether prompting and confidence calibration let them answer NOT FOUND instead of guessing when reading fields from blurred, poorly framed, or badly lit photographs, though, has not, to the author's knowledge, been systematically measured. This proposal asks which combination of prompting strategy and confidence-calibration method gives small open-weight VLMs the best accuracy-abstention-latency trade-off when reading requested fields from product photographs. It also asks whether that combination stays reliable as image quality falls. Three controlled computational studies, using authentic blind-user photographs, clean text-reading images, and a custom set of Bangladeshi product labels, compare prompting strategies, confidence signals under graded degradation, and deployment configurations on GPU, CPU, edge-runtime, and cloud targets. Together they will yield a decision matrix of the best model, prompt, confidence method, threshold, and precision within a latency budget, as evidence for low-cost assistive-device design. Human evaluation with blind and low-vision users is the planned next phase.*

I. Research Problem

In 2020, an estimated 43.3 million people worldwide were blind, a figure projected to rise to 61 million by 2050 [1]. Roughly 90% of people with vision loss live in low- and middle-income countries (LMICs) [2], including Bangladesh. There, imported assistive devices and subscription-based sighted-assistance services can remain out of reach for many users. Camera-based artificial intelligence (AI) assistants built on vision-language models (VLMs), such as Be My AI and Doubao, now offer a cheaper alternative: point a camera and ask what is in front of you. But these systems typically depend on a live cloud connection, and the models behind them can answer confidently even when the image does not support an answer.

The images blind users actually capture make this worse. They are frequently blurred, off-center, or poorly lit, and about 28% of the questions blind users ask cannot be answered from the photograph at all [3]. Under exactly these conditions, general-purpose VLMs tend to hallucinate incorrect visual details rather than admit they cannot tell [4]. The failure also grows sharply as image quality falls. On 1,859 blind-user product photographs, GPT-4.1 fell from 98.5% to 74.9% accuracy once blur, poor framing, or rotation was present, while two large open models run at 4-bit precision (Llama 3.2 90B and Molmo 72B) reached only 44.1% and 36.1% [5]. Recent preprint evidence suggests that 4-bit quantization costs little accuracy but lowers the internal confidence signal [6].

Whether any prompting strategy or confidence-calibration method can recover a reliable "I don't know" under this joint pressure of image degradation and quantization has not, to the author's knowledge, yet been systematically measured. Settling it takes controlled, comparative measurement across models, prompts, and calibration methods. That measurement, not a working device, is the contribution this study sets out to make.

II. Existing Literature

Blind and low-vision (BLV) people already ask vision-language models (VLMs) about everyday objects, and the studies of this practice report recurring problems. VizWiz, built from questions that blind people asked about their own photographs, showed that such images are often blurred, poorly framed, or badly lit, and that about 28% of questions cannot be answered from the image [3]. A formative study with 15 visually impaired participants named low latency, low cognitive load, and hallucination-resistant answers as core requirements [7]. Blind users typically heard responses about ten times longer than their own input and turned to prompt-based workarounds [8]. Even so, BLV raters favored long-form answers overall, and preferred shorter ones only for questions about a specific attribute [4]. The same raters trusted model answers more than sighted raters did, although the models often hallucinated visual details, especially for blurry images [4]. Blind users also struggle to verify AI output non-visually [9].

Failures concentrate where capture conditions are poor. Chiu et al. labeled 39,181 blind-user images with six quality flaws and noted that synthetic distortions lack the complexity of natural ones [10]. On 1,859 blind-user product photographs, GPT-4.1 fell from 98.5% to 74.9% accuracy when blur, poor framing, or rotation was present, and two large open models run at 4-bit precision (Llama 3.2 90B and Molmo 72B) reached 44.1% and 36.1% [5]. Models also fail on small evidence regions and text recognition [11]. For Bengali, BanglaWild benchmarks scene-text transcription across fifteen VLMs [12]. Transcription, though, is not field extraction with refusal under blind-user capture.

Abstention is the standard remedy. Temperature scaling recalibrates overconfident models [13], and selective classification trades coverage against risk [14]. In visual question answering (VQA), thresholding softmax scores limited models to answering

under 8% of questions at 1% risk [15]. Likelihood- and sampling-based confidence on VizWiz-VQA showed no clear winner [16]. Verbalized confidence in text-only models tends to be overconfident, while agreement among sampled answers, an idea from self-consistency decoding [17], improves failure prediction without a consistent winner [18]. For BLV users, an explicit abstention instruction increased abstention on unanswerable questions, at some cost in precision [4].

On-device use favors small quantized models. SmolVLM's smallest variant (SmolVLM-256M) needs under 1 GB of GPU memory [19], and 4-bit precision is near-optimal for accuracy per memory bit in large language models (LLMs) [20]. Recent preprints suggest that confidence degrades faster than accuracy: quantizing Qwen2-VL to 4-bit cost 1.6 accuracy points but lowered internal error-detection area under the receiver operating characteristic curve (AUROC) from 0.95 to 0.80 [6], and in a five-attempt pilot SmolVLM-Instruct produced a parseable verbalized confidence only once [21].

Research gap. So far as the author can determine, these threads have not been joined. The closest work compares confidence signals in small VLMs under synthetic degradation, but it does so on 100-image multiple-choice food photographs that force an answer and cannot observe refusal [6], [21]. Blind-user studies document the need for concise, reliable answers [4], [7]–[9], and one compares abstention prompts [4], though not on small quantized models under graded degradation. A literature search (September 2026) found no study that combines open-ended field extraction with an explicit not-found option, authentic blind-user images alongside graded corruption, small 4-bit open models, and Bengali-market packaging. Graded synthetic corruption offers control [22], [23], but it cannot replace authentic captures [5], [10]. This study uses both.

III. Research Question

Central research question.

> *Which combination of prompting strategy and confidence-calibration method gives small open-weight vision-language models (VLMs), run across 16-bit and 4-bit configurations, the best accuracy-abstention-latency trade-off when reading requested fields from product photographs, and does that combination remain reliable as image quality degrades to the level typical of blind-user photography?*

The question is tested on authentic blind-user photographs, which carry natural quality flaws. It is also tested on clean text-reading images and a custom set of Bangladeshi product labels, both degraded at three controlled severities from clean (L0) to severe (L2). "Best" means the configuration (model, prompt, confidence method, threshold, and precision) that maximizes accuracy within a latency budget. Three sub-questions, one per study, divide the problem:

RQ1 (prompting strategy). How do untargeted description, the natural question (the dataset's own question, defined per dataset in Section IV), and targeted extraction (one named field, with a NOT FOUND option) differ in extraction accuracy, hallucination rate, output length, and latency?

RQ2 (calibration under degradation). Which confidence signal (internal token probability, verbalized confidence, or self-consistency) best separates correct from incorrect answers as image quality degrades, as measured by error-detection AUROC and coverage at fixed accuracy? Does temperature scaling improve expected calibration error (ECE) and the transfer of thresholds fitted on clean images to degraded ones?

RQ3 (deployment trade-off). What accuracy-latency-memory frontier do model and precision configurations occupy on GPU, CPU, and edge-runtime targets, and how does a free-tier cloud vision API compare?

IV. Methodology

This study is a controlled, computational comparative experiment. It uses no human participants and no paid services, and all experiments run on a laptop and on the free tier of a cloud notebook service with a single 16 GB GPU (NVIDIA T4). The design has one dataset-construction stage and three studies, each answering one sub-question from Section III. Fig. 1 shows the pipeline.

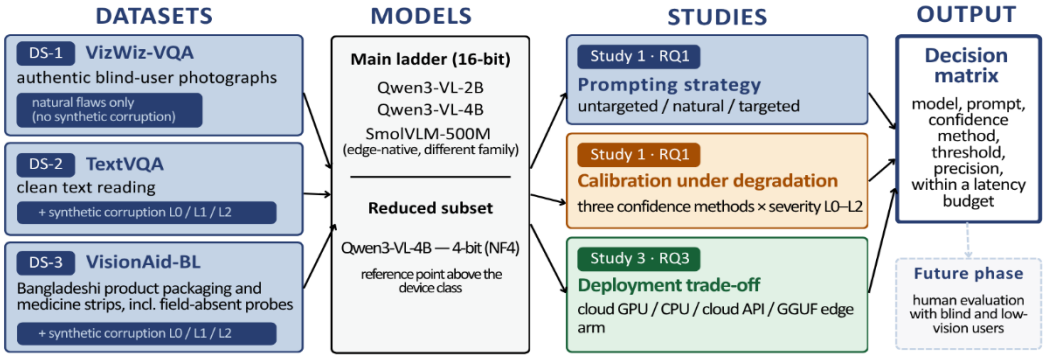


Fig. 1. Overview of the study design. Three datasets (authentic blind-user photographs, clean text-reading images, and a custom set of Bangladeshi product labels) are processed by small open-weight vision-language models in three linked studies on prompting strategy, confidence calibration under degradation, and deployment trade-offs. Their results combine into a decision matrix for device design. Human evaluation is planned as a later phase (dashed).

A. Datasets

Three datasets play distinct roles.

DS-1: VizWiz-VQA (validation split). VizWiz-VQA contains 4,319 validation questions about photographs taken by blind users, each with ten crowd-sourced answers [3]. The dataset is licensed CC BY 4.0. Its images carry the real capture failures of this population: blur, poor framing, and poor lighting. Roughly 28% of its questions are unanswerable under a majority-vote definition [3]. These are deliberately *retained* for Study 2, because they are the cases where a system should abstain. A reading subset (short-text answers such as dates, labels, and prices) is built by keyword filtering and checked by hand on 50 random items. Study 1, by contrast, uses only answerable items, since extraction accuracy is undefined otherwise. VizWiz-QualityIssues labels [10] are used to stratify results by authentic flaw type, such as blur or framing.

DS-2: TextVQA (validation split). A sample of about 300 questions from TextVQA [24] provides sharp images with clean ground truth. Comparing DS-2 with DS-1 separates "the model cannot read" from "the model cannot cope with blind-user capture conditions."

DS-3: VisionAid-BL (custom). About 150 items of Bangladeshi product packaging and medicine strips are captured with a head-mounted webcam. Each item is photographed under two conditions, best-effort (steady, well lit) and *simulated* blind-user capture (motion blur, oblique angle, partial occlusion, low light), for roughly 300 images in total. The simulated condition is reported as simulated, not as real blind-user capture. Correct field values are recorded at capture time. About 20–25% of items are *field-absent probes* (for example, asking for an expiry date on a package face that has none), so that "NOT FOUND" is the correct answer and over-abstention can be measured.

Controlled corruption. To make degradation a formal independent variable, synthetic corruptions are applied to DS-2 and DS-3 but not to DS-1, which keeps its natural flaws. Three severities are used: L0 (clean), L1 (mild), and L2 (severe). The types are Gaussian blur, motion blur, brightness reduction, crop/framing shift, rotation, and occlusion. Parameters are fixed before any evaluation and validated on a pilot. Accuracy must fall monotonically from L0 to L2; if it does not, the parameters are revised before locking.

B. Models and Decoding

Every model is open-weight and pinned by revision, with library versions recorded. The design uses a scale ladder within one current family, the Qwen3-VL series (2B and 4B parameters at 16-bit precision), plus one edge-native model from a different family, SmolVLM-500M [19]. A reduced subset also runs Qwen3-VL-4B at 4-bit precision in the NormalFloat (NF4) format, as a reference point above the device class. Keeping the ladder within a single family isolates the effects of scale and quantization from architecture, while the second family guards against results specific to one vendor. Extended "thinking" output is disabled for all models. Main runs use greedy decoding, and sampling appears only inside the self-consistency method.

C. Study 1: Prompting Strategy (RQ1)

All three datasets go through Study 1. Every selected image is processed under three prompts, which makes this a within-items design in which each image is its own control. The prompts are (a) *untargeted* ("Describe all text in this image"); (b) *natural*, meaning the dataset's own question, unconstrained: the user's original VizWiz question for DS-1, the original TextVQA question for DS-2, and the field request recorded at capture for DS-3; and (c) *targeted* ("Extract only [field]; answer with the value or NOT FOUND"). Token limit and decoding settings are identical across prompts.

The dependent variables are field-extraction accuracy (VQA soft accuracy and normalized exact match, with average normalized Levenshtein similarity (ANLS) for text fields; rules for dates and prices are fixed in advance); *hallucination rate*, defined identically in all conditions as asserting a wrong field value, or any value when the field is absent; output length in words; and latency. For untargeted output, correctness is scored by whether the output contains the correct value. Abstention outputs are scored separately from answers, so a model cannot gain credit by outputting the word "unanswerable." Paired differences are tested with mixed-effects models with item as a random effect, and effect sizes are reported alongside confidence intervals.

D. Study 2: Calibration Under Degradation (RQ2)

The question driving Study 2 is which confidence signal best identifies wrong answers when input quality falls. Every model output receives a confidence score. A threshold τ then decides whether the system speaks or abstains, as in Fig. 2. The design is a repeated-measures factorial: model × confidence method × corruption severity (L0, L1, L2).

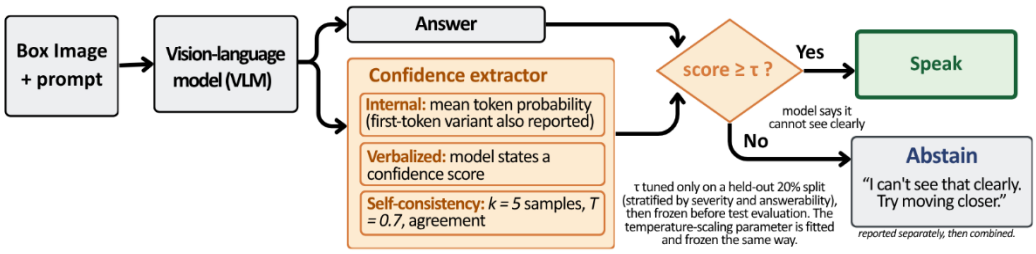


Fig. 2. Confidence-gated abstention. Each model answer receives a confidence score from one of three methods. A threshold τ, tuned on a held-out split and then frozen, decides whether the system speaks or abstains. The inset sketches, for illustration only, how error among answered items is expected to fall as coverage shrinks; it does not show results.

Three confidence methods are compared:

Internal confidence (baseline): mean token probability of the generated answer. A first-token variant is also reported, because aggregation choice has been shown to matter for extraction [23].

Verbalized confidence: the model states a confidence score and may declare that it cannot see clearly [18]. Its output-parse rate is reported as a result in itself, since prior work found it can collapse under 4-bit quantization [6].

Self-consistency: k = 5 sampled answers at temperature 0.7; agreement among them is the score [17]. The extra compute is included in the latency figures.

Recalibration by temperature scaling [13] is treated separately. A monotone rescaling cannot change how items are ranked, so by construction it cannot alter error-detection AUROC or the risk-coverage curve. It is therefore evaluated only on what it can change: expected calibration error (ECE) [13] and *threshold transfer*, meaning how well a threshold fitted on clean images performs on L1 and L2 images. The scaling parameter is fitted on the same held-out split and frozen before test evaluation.

Ground truth for "should abstain" is defined in two independent ways. In the first, the model's answer is wrong; in the second, the human annotators marked the question unanswerable. *Primary metrics* are error-detection AUROC (the probability that a correct answer is scored above an incorrect one; 0.5 is chance) and coverage at fixed accuracy (the largest fraction of questions that can be answered while meeting a target accuracy set in advance from the device specification) [16]. Secondary metrics are the area under the risk-coverage curve [14], ECE, and abstention precision. Model-stated abstention and score-below-threshold abstention are reported separately and then combined.

Rigor controls. Thresholds are tuned only on a held-out 20% split, stratified by corruption severity and answerability, and then *frozen before test evaluation*. Random seeds are fixed. All intervals are bootstrap confidence intervals resampled by item. Results are pooled across severities for the main claims and also reported per severity, so the study can say whether each method degrades gracefully or collapses. Sample sizes target at least 300 items per model and condition on DS-1 and DS-2 combined, since about 100 items yields an AUROC interval of roughly ±0.09. DS-3 is reported descriptively as an exploratory set.

E. Study 3: Deployment Trade-Off (RQ3)

Four deployment targets anchor Study 3's benchmark for each model and precision: a cloud GPU (T4), CPU only, a free-tier cloud vision API, and an edge-runtime arm that runs SmolVLM-500M through llama.cpp in GGUF (GPT-Generated Unified Format), a runtime closer to real devices. Recorded measures are median and 95th-percentile latency, peak memory, and accuracy carried over from Studies 1 and 2. The cloud arm uses only DS-3 and TextVQA, not VizWiz, to avoid sending blind users' photographs to a third party. Accuracy, latency, and memory are combined into a Pareto frontier. Results are reported as relative comparisons.

TABLE I

Summary of the three studies: design, independent variables (IVs), and primary dependent variables (DVs)

Study	Design	Independent variables	Primary dependent variables
1: Prompting strategy (RQ1)	Within-items comparison; each image processed under all three prompts	Prompt (untargeted, natural, targeted) × model	Field-extraction accuracy; hallucination rate; output length (words); latency
2: Calibration under degradation (RQ2)	Repeated-measures factorial	Model × confidence method (internal, verbalized, self-consistency) × severity (L0, L1, L2)	Error-detection AUROC; coverage at fixed accuracy. Secondary: area under the risk–coverage curve, ECE, abstention precision
3: Deployment trade-off (RQ3)	Factorial benchmark	Model × precision × target (cloud GPU, CPU only, free-tier cloud API, GGUF edge arm for SmolVLM-500M)	Latency (median and 95th percentile); peak memory; accuracy carried over from Studies 1 and 2; Pareto frontier

AUROC = area under the receiver operating characteristic curve; ECE = expected calibration error; GGUF = GPT-Generated Unified Format; L0/L1/L2 = clean/mild/severe corruption; the cloud API arm uses DS-2 and DS-3 only.

F. Integration and Limitations

Together, the three studies produce a decision matrix that gives, for each assistive task, the model, prompt, confidence method, threshold, and precision that maximize accuracy within a latency budget. The budget is set from the device specification in advance and checked by sensitivity analysis. That turns the findings into design requirements for a low-cost assistive device.

Limitations are stated in advance. There is no human evaluation in this phase, and usability and trust testing with blind and low-vision users is the planned next phase. DS-3 uses simulated capture failures, and synthetic corruption on DS-2 and DS-3 approximates, but does not reproduce, real capture failures [5], [10]. The GPU and CPU are hardware proxies, and public benchmarks may overlap with model training data. Models in the reduced subset are tested on fewer items. Prior work has already examined calibration on VizWiz [16] and the effect of quantization on confidence signals under degradation [6]. This study extends both to small quantized models, open-ended extraction with refusal, authentic blind-user images, and Bengali-relevant packaging labels.

V. Conclusion

Small, quantized, open-weight VLMs are a realistic route to affordable, offline assistive reading for blind and low-vision people. Whether they can answer NOT FOUND instead of guessing when a photograph is blurry, off-center, or poorly lit has not, as far as the author can tell, been systematically measured. This proposal goes after that gap with three linked computational studies, run on authentic blind-user photographs, degraded text-reading images, and a custom set of Bangladeshi product labels. They cover prompting strategy (Study 1), confidence calibration under degradation (Study 2), and the accuracy-latency-memory trade-off (Study 3).

The question is urgent. Blindness is projected to reach 61 million people by 2050 [1], concentrated in low- and middle-income countries such as Bangladesh [2], so the affordable route to assistive AI increasingly runs through small, offline models rather than paid cloud services. Yet poor image quality already lowers accuracy [5], and preprint evidence suggests that 4-bit quantization weakens a model's ability to tell when it is wrong [6]. Whether that trade-off can be managed, or quietly makes assistive AI less trustworthy for the users who can least afford a wrong answer, is still open.

It is also new. To the best of the author's knowledge, no earlier study combines open-ended field extraction with abstention, authentic blind-user images, small quantized open models, and Bengali-market packaging. The closest work tests confidence signals on a 100-image multiple-choice food-photo task that forces an answer [6], [21]. Here, all three studies run on the same blind-user and Bangladeshi-label images, so the result is one decision matrix rather than three separate findings. It names the model, prompt, confidence method, threshold, and precision that perform best within a latency budget, and it settles choices still open in a device-level specification for an offline assistive wearable: which model, which prompt, and which abstention threshold to deploy.

The design has stated limits: no human evaluation in this phase, partly simulated blind-user capture, GPU and CPU as proxies for phone-class hardware, reduced-subset models tested on fewer items, and possible benchmark overlap with model training data. Testing the chosen configuration with blind and low-vision users is therefore the planned next step. The goal is for developers in low- and middle-income countries such as Bangladesh to build assistive system that admit uncertainty rather than mislead.

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